EnergyFlux

AI-Powered Energy Intelligence Platform

Physics simulation generates real-time energy data.
AI agents understand, diagnose, and optimize the system.

Bifacial Solar PV	Battery Storage	Grid Simulations	AI Diagnosis Agent
185 kWp	**400 kWh**	**17,518**	**NVIDIA NIM**
pvlib physics model	TOU rate optimization	pandapower NR solver	Nemotron-Ultra + RAG

Solar Forecasting → PV Modeling → BESS Dispatch → Power Flow → AI Diagnosis

Chennan Li, PhD, PE

System Architecture — AI Integrated at Every Layer

PHYSICS ENGINE

Simulation & Modeling

pvlib

PV cell-level modeling
4 configurations compared

pandapower

Newton-Raphson power flow
17,518 hourly grid solutions

BESS Model

SOC dynamics, efficiency
TOU rate-aware dispatch

DATA PIPELINE

Collection & Processing

Open-Meteo API

2-year weather history
GHI, DNI, temperature, wind

Time-Series DB

Hourly PV output, load
bus voltages, line loading

Knowledge Base

Engineering docs, specs
Fault patterns, O&M guides

AI / ML LAYER

Intelligence & Decision

NHITS Neural Net

Solar irradiance forecast
24-hour ahead prediction

NVIDIA Nemotron

ReAct reasoning agent
Autonomous tool calling

FAISS RAG

Vector similarity search
Grounded diagnosis output

Data Flow: Weather → Forecast (AI) → PV Generation (Physics) → BESS Dispatch → Power Flow (Physics) → Anomaly Detection → AI Diagnosis (NVIDIA NIM)

Simulation Results — Power Flow + BESS + Voltage Analysis

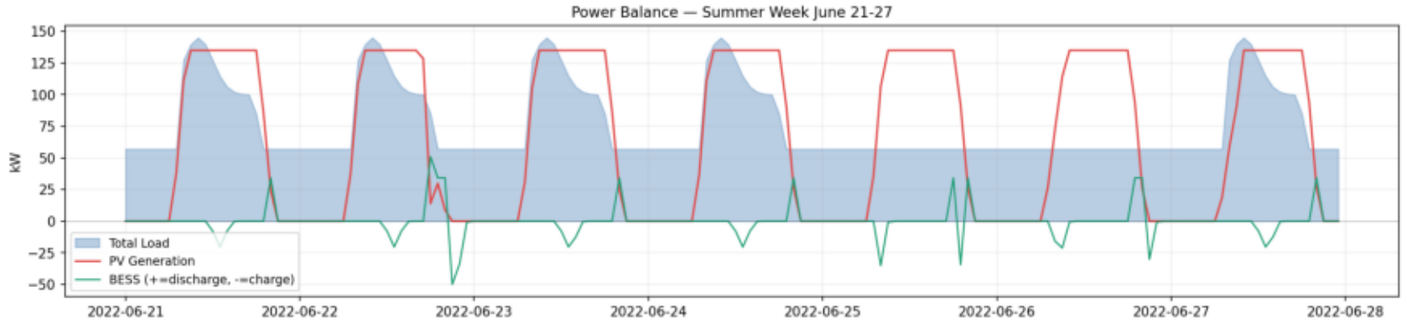
5,363 hrs overvoltage (31%)

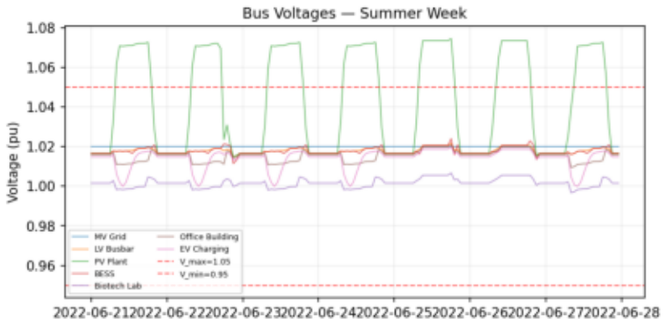
Peak: 1.074 pu (limit 1.05)

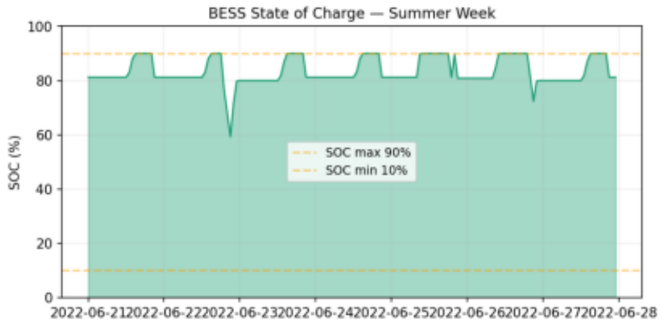
BESS: 239 cycles, TOU arbitrage

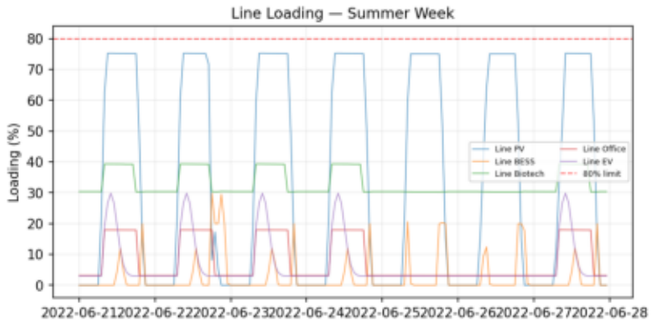
Lines: max 75% loading (safe)

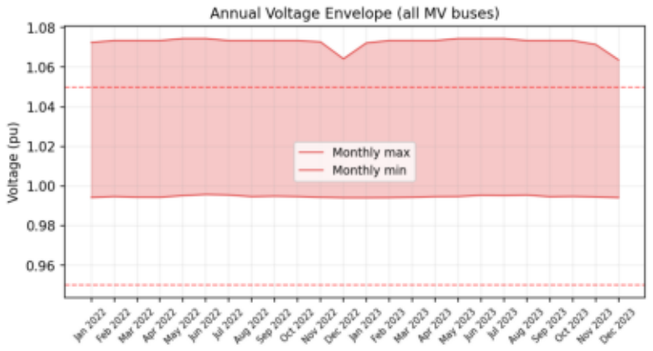
Industrial Park Power Flow — Fremont CA | PV=187kWp | BESS=100kW/400kWh | 10kV Distribution











AI Agent in Action — Powered by NVIDIA NIM (Nemotron-Ultra-253B)

Autonomous Root Cause Analysis

The agent receives an alert, decides what to investigate, calls tools, and produces a structured diagnosis — no human in the loop.

TRIGGER: Overvoltage detected — PV bus 1.068 pu > 1.05 limit | BESS SOC 90%

1 `query_weather()` — Clear sky, GHI 930 W/m²

Max solar irradiance → PV at full capacity

2 `query_pv_status()` — PV = 135 kW, Load = 57 kW

78 kW surplus with nowhere to go

3 `query_powerflow()` — PV bus 1.068 pu, reverse flow

Surplus pushes through 300m cable → voltage rise

4 `query_knowledge_base()` — RAG: overvoltage pattern match

Historical: midday + clear + full BESS = classic scenario

AI OUTPUT (structured JSON)

Root Cause: PV surplus (78kW) + saturated BESS → uncontrolled reverse power flow → voltage rise

Actions: ① Increase BESS to 200kW/800kWh ② Enable inverter VAR support ③ Shift EV charging to midday

NVIDIA Technology Stack

NVIDIA NIM	API endpoint for Nemotron-Ultra inference	**Cloud or on-prem deployment ready**
Nemotron-Ultra 253B	Tool-calling LLM ReAct reasoning loop	**Decides WHAT to investigate and WHY**
LangChain ReAct Agent	Orchestrates multi-step tool calls autonomously	**No predefined rules — LLM decides**
FAISS RAG	Vector search over engineering knowledge base	**Grounds answers in real documents**

Why NVIDIA NIM?

Enterprise-ready · On-prem deployable · No OpenAI dependency · GPU-optimized inferen

AI Expansion Roadmap — From Diagnosis to Autonomous Optimization

TODAY: Single Agent — Reactive Diagnosis

Alert triggers agent → Agent investigates autonomously → Structured diagnosis + recommended actions

NVIDIA NIM · Nemotron-Ultra · 4 tools · RAG knowledge base · JSON output for downstream automation

NEXT: Multi-Agent System — Proactive Optimization

Forecast Agent

Predicts tomorrow's PV output
+ load demand using weather forecast

Pre-positions BESS SOC for expected surplus/deficit

↔

Dispatch Agent

Optimizes BESS charge/discharge schedule against TOU electricity rates

Pyomo LP solver + LLM for constraint interpretation

↔

Grid Agent

Monitors voltage and loading in real-time across all buses

Triggers corrective action before limits are reached

↔

Coordinator

Orchestrates all agents, resolves conflicts between objectives

Multi-agent consensus via NVIDIA NIM inference

VISION: Autonomous Energy Management

Physics simulation provides the ground truth. AI agents provide the intelligence.
The platform evolves from reactive (diagnose after alert) to predictive (forecast + pre-position)
to autonomous (optimize in real-time without human intervention).
All inference powered by NVIDIA NIM — deployable on-premise for industrial data privacy.

Powered by: NVIDIA NIM · Nemotron-Ultra · LangChain · FAISS · pvlib · pandapower · Pyomo · FastAPI